REAL-TIME WEATHER DATA PIPELINE

AWS · Databricks · Apache Airflow · Medallion Architecture

127K+
Records

53
Columns

11
DAG Tasks

4
AWS Layers

PROJECT TEAM

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PROBLEM STATEMENT & PROJECT OVERVIEW

Why this pipeline was built — and what it delivers

PROBLEM STATEMENT

Traditional weather data pipelines rely on manual batch uploads, scheduled scripts, and fragile ETL jobs that break under scale, offer no real-time visibility, and leave data engineers with no audit trail when something fails. There was no automated, event-driven system to ingest, clean, and model global weather data into analytics-ready tables — reliably, at scale, with full observability.

PROJECT OBJECTIVE

To build an end-to-end real-time weather data pipeline that automatically ingests raw CSV data, processes it through a Medallion Architecture (Bronze → Silver → Gold), and delivers clean, analytics-ready Star Schema tables using AWS and Databricks — with full orchestration, monitoring, and zero manual intervention.

The pipeline reflects real-world, production-grade data engineering — event-driven ingestion via AWS Lambda and Amazon Kinesis, structured transformation using PySpark on Databricks Serverless, dimensional modelling in the Gold layer, end-to-end orchestration via Apache Airflow on AWS MWAA, and real-time Slack alerting on every pipeline task. Every layer includes validation checkpoints and a full audit trail ensuring every record is traceable from source CSV to the final Fact table.

PROJECT OVERVIEW

What we built, the approach we took, and the value delivered

The Dataset

GlobalWeatherRepository CSV — 127,646 rows × 53 columns covering temperature, humidity, wind, air quality, UV index, and astronomical data. Zero nulls, zero duplicates at source.

Our Approach

Medallion Architecture (Bronze → Silver → Gold) on Databricks with AWS S3 as the storage backbone — the industry standard for modern, governed data lakes at enterprise scale.

Real-Time Ingestion

CSV upload to S3 triggers Lambda → Kinesis Data Stream → Kinesis Firehose → S3 Bronze (Parquet). Fully automated, event-driven — zero manual intervention at any stage.

Business Outcome

Clean Star Schema in Gold layer: 1 Fact + 4 Dim tables. Databricks dashboard with weather insights across temperature trends, air quality, UV risk, and global comparisons.

TECHNOLOGY STACK

Every service, every tool — and exactly why it was chosen

STORAGE

AWS S3

TRIGGER

AWS Lambda

STREAMING

Kinesis Data Stream

DELIVERY

Kinesis Firehose

CATALOG

AWS Glue + Crawler

COMPUTE

Databricks Serverless

PROCESSING

Apache Spark / PySpark

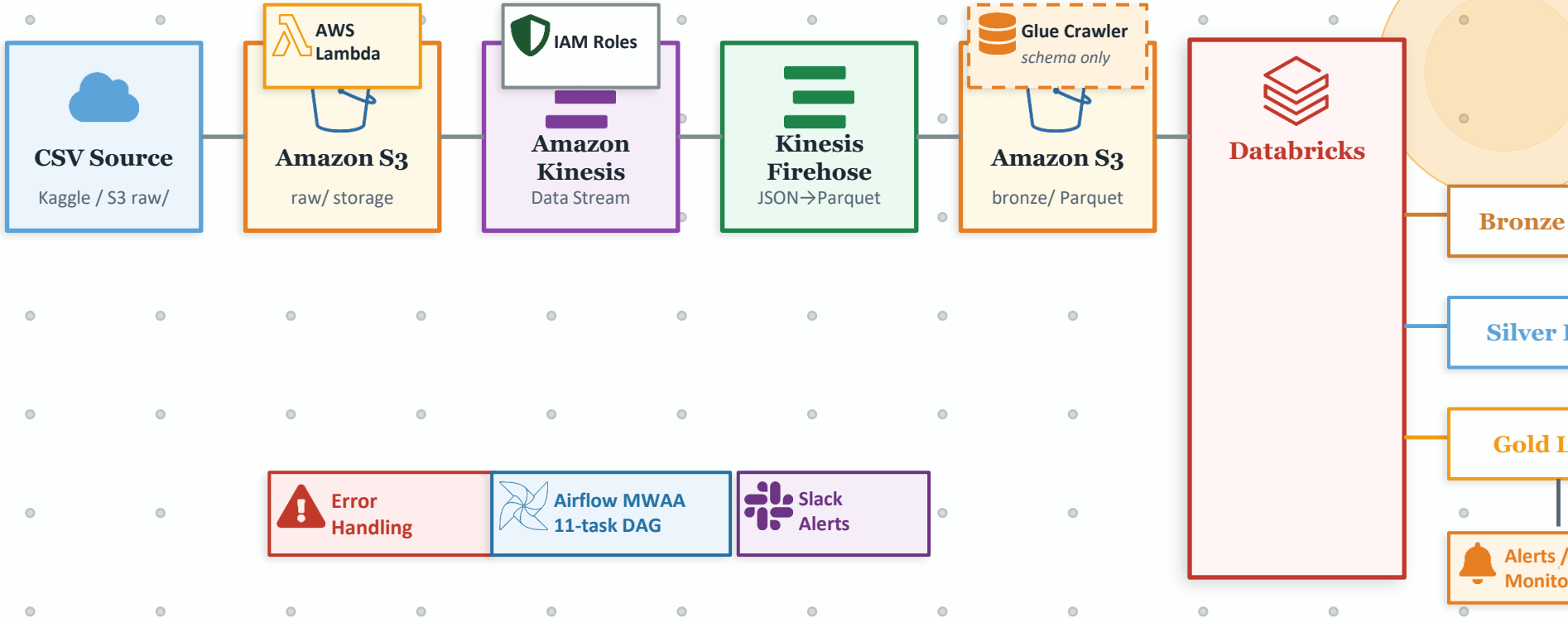
ORCHESTRATION

Airflow (AWS MWAA)

ALERTING

Slack Webhook

HIGH LEVEL DESIGN



PIPELINE ARCHITECTURE

Complete end-to-end data flow — ingestion to analytics

CSV Source
GlobalWeather Repository

S3 raw/
S3 Event Notification

AWS Lambda
~256 batches of 500 rows

Kinesis Data Stream
On-Demand ap-south-1

Kinesis Firehose
JSON→Parquet bronze/

S3 bronze/
Parquet · immutable partitioned by country

Glue Crawler
schema metadata only

S3 bronze/
reads directly

Databricks Silver
Clean & Transform

Databricks Gold
Dim/Fact Star Schema

Airflow MWAA
11-task DAG Orchestration

Slack Alerts
Real-time Notifications

127,646	~256	54 cols	4	11
Records Streamed	Kinesis Batches	Glue Schema	IAM Roles	Airflow Tasks

DATASET PROFILE & S3 SETUP

Understanding our data and cloud storage foundation

DATASET PROFILE

Rows	127,646
Columns	53 + 1 PK
Null Values	16000
Duplicate Rows	600
Primary Key	weather_record_id
Format	CSV → Parquet
Partition Key	Country
Region	ap-south-1 (Mumbai)
Bucket	global-weather-pipeline

S3 BUCKET STRUCTURE

s3://global-weather-pipeline/

├─ raw/	CSV upload → triggers Lambda
├─ bronze/	Firehose writes Parquet automatically
├─ silver/	Databricks Silver writes here
├─ gold/	Dim + Fact tables (final output)
├─ scripts/	Lambda + Glue code references
├─ airflow/dags/	MWAA DAG Python file

BRONZE LAYER — Raw Ingestion

Store data exactly as it arrived · The immutable source of truth

WHAT BRONZE DOES

01 Raw CSV Ingestion

Reads from S3 raw/ with inferSchema — no casting, no transformations.

02 3 Audit Columns Only

_bronze_ingested_at · _bronze_source_file · _bronze_batch_id — only permitted additions.

03 Parquet Write to S3

Writes to bronze/ partitioned by Country — enables partition pruning downstream.

04 Hive Metastore Table

Registered as: SELECT * FROM weather_bronze.raw_weather_data

05 Validation Report

Row count (127,646) · column count (57) · partition existence check.

06 Firehose + Glue Assist

Format conversion handled upstream — Bronze notebook is clean and lightweight.

DESIGN DECISIONS & AUDIT

Q Why keep weather_record_id in Bronze?

Carried from source — avoids regeneration in Silver, preserves end-to-end lineage from CSV to Gold Fact table.

Q Why Parquet and not CSV?

3–5× storage savings. Firehose converts automatically via Glue schema. 10× faster columnar reads in Silver.

Q Why partition by Country?

Spark partition pruning — Silver reads only the needed country folder, not all 127K rows.

Q What do Audit Columns achieve?

_ingested_at tracks arrival time. _source_file provides lineage. _batch_id links rows to specific Lambda invocation.

Q Where does Audit happen in Bronze?

Validation cell runs post-write: row count = 127,646, column count = 57 (53+1+3), partitions exist in S3.

SILVER LAYER — Data Cleaning & Transformation

Make data trustworthy and consistent — where the real engineering happens

STEP 01 Column Rename & Schema Enforcement

- air_quality_PM2.5 → air_quality_pm2_5 (dots not allowed in Glue/Spark)
- air_quality_us-epa-index → air_quality_us_epa_index
- Strict schema: DoubleType, IntegerType, LongType, TimestampType
- Enforces contract between Bronze and Gold — Gold trusts Silver completely

STEP 02 Type Casting & Standardization

- last_updated string → TimestampType (to_timestamp function)
- temperature_celsius → DoubleType | humidity → IntegerType
- last_updated_epoch → LongType | wind_kph → DoubleType
- Condition text case standardized ('Partly cloudy' vs 'Partly Cloudy')

STEP 03 Missing Value Imputation

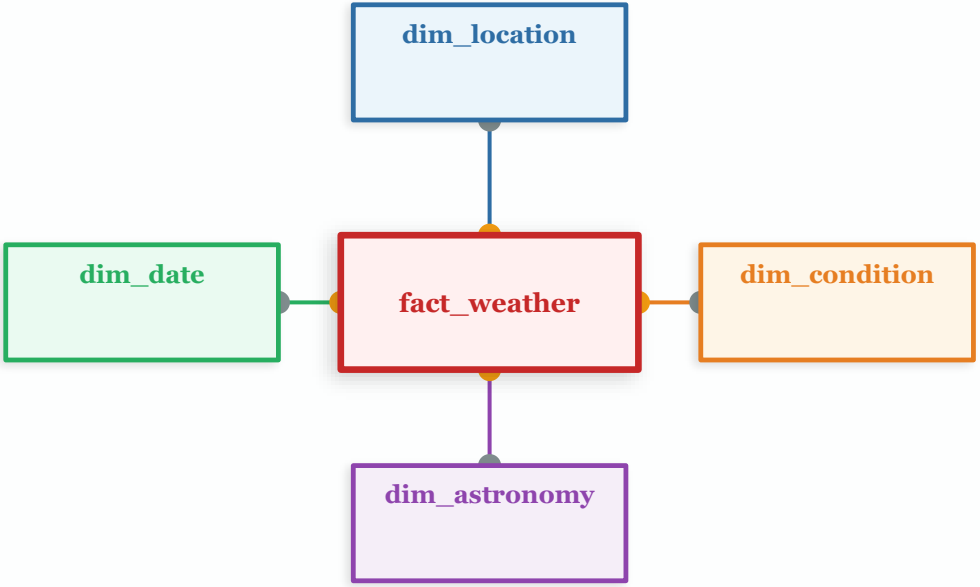
- Null temperature_celsius → filled with column mean (non-null rows only)
- Null humidity → int(mean) | Null wind_kph → float(mean)
- Null air_quality_pm2_5 → float(mean) | Null condition_text → 'Unknown'
- Means computed BEFORE imputation to avoid bias — production best practice

STEP 04 Deduplication & Outlier Removal

- dropDuplicates() on full row — verified against Bronze count
- Filter humidity > 100 → physically impossible sensor reading
- Filter temperature_celsius = 999 → known sensor error code
- Filter uv_index < 0 → physically invalid, 650 rows detected & removed

GOLD LAYER — Dimensional Modelling

Star Schema: 4 Dimension Tables + 1 Fact Table · Business-ready analytics



STAR SCHEMA ERD

FACT TABLE — fact_weather

· weather_record_id (PK) · FK → location_id, condition_id, astronomy_id, date_id

· temperature_celsius / fahrenheit · wind_kph / mph · pressure_mb · humidity

· uv_index · cloud · visibility_km · feels_like · gust_kph · precip_mm

· air_quality: pm2_5, pm10, carbon_monoxide, ozone, no2, so2, epa_index, defra_index

dim_location

· location_id (PK) · country, location_name

· latitude, longitude · timezone

· region_category · weather_station · sensor_type

dim_condition

· condition_id (PK) · condition_text

· temperature_category · humidity_level

· wind_intensity · precipitation · uv_risk · air_quality

dim_astronomy

· astronomy_id (PK) · sunrise / sunset

· moonrise / moonset · moon_phase

· moon_illumination

dim_date

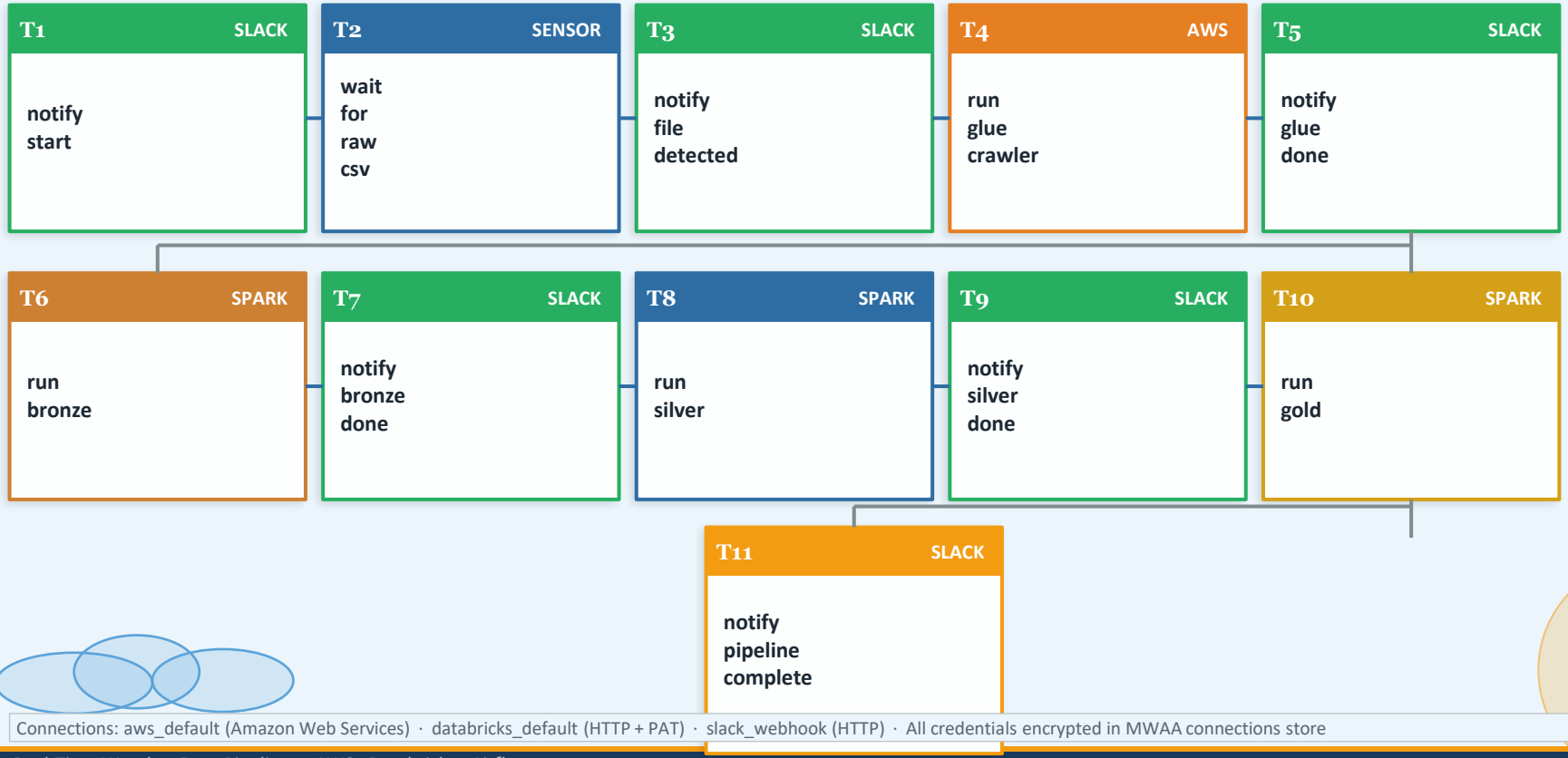
· date_id (PK) · date · hour · month · year

· day_of_week · quarter · season

· is_weekend (derived from last_updated)

AIRFLOW ORCHESTRATION — AWS MWAA

11-task DAG · global_weather_pipeline · Full observability via Slack alerts



Connections: aws_default (Amazon Web Services) · databricks_default (HTTP + PAT) · slack_webhook (HTTP) · All credentials encrypted in MWAA connections store

ERROR HANDLING · DATA QUALITY · ALERTS & MONITORING

Pipeline resilience — how we detect, handle, and alert on every failure

ERROR HANDLING

✗ Lambda OOM/timeout → AWS auto-retries up to 2× · CloudWatch logs

✗ Kinesis throttling → boto3 retry logic handles automatically

✗ Notebook failure → Airflow marks FAILED → downstream SKIPPED

✗ Any task failure → Slack immediate alert with task name + error

✗ Recovery: re-run failed task only — Bronze is immutable, no data loss

DATA QUALITY CHECKS

Bronze	✓ Row count = 127,646 · Column count = 57 · Partitions exist
Silver	✓ Zero nulls after imputation · Dedup count logged · Outlier counts
Silver	✓ humidity>100 removed · temp=999 removed · UV<0 removed (650 rows)
Gold	✓ Fact row count = Silver count · FK completeness · Dim PK uniqueness
Audit	✓ Full trace: weather_record_id source CSV → Gold Fact table

ALERTS / MONITORING / LOGGING

Slack Alerts

· 11 Slack notifications — every task state change

· Pipeline start · each layer complete · success or failure

· Immediate failure alert with task name + timestamp

· Channel: #weather-pipeline (MWAA HTTP connection)

Airflow Monitoring

· MWAA UI: DAG run history · task status per run

· Task instance logs with full stdout/stderr

· Retry tracking · skip tracking · execution duration

· CloudWatch logs for all MWAA components

Logging & Lineage

· Error logs table in Databricks (error_logging notebook)

· AWS CloudTrail: every IAM API call logged + timestamped

· Glue Crawler run history: schema change log per crawl

· 3 Bronze audit cols: arrival time · source file · batch ID

DASHBOARDS & BUSINESS INSIGHTS

Analytics built on the Gold Star Schema — actionable weather intelligence

Temperature Trends

Average temperature by country and region — identifies hottest zones. Celsius/Fahrenheit trends across seasons and time of day.

Wind Speed Analysis

Wind intensity level distribution by location. Peak gust speeds, dominant wind directions, and calm vs storm frequency analysis.

Precipitation Patterns

Precipitation level categories by country. Humidity vs precipitation correlation. Rain frequency and heavy rainfall hotspot identification.

UV Risk Index

UV risk level distribution globally — Low/Moderate/High/Very High. Peak UV hours by timezone. Countries with highest UV exposure.

Air Quality Dashboard

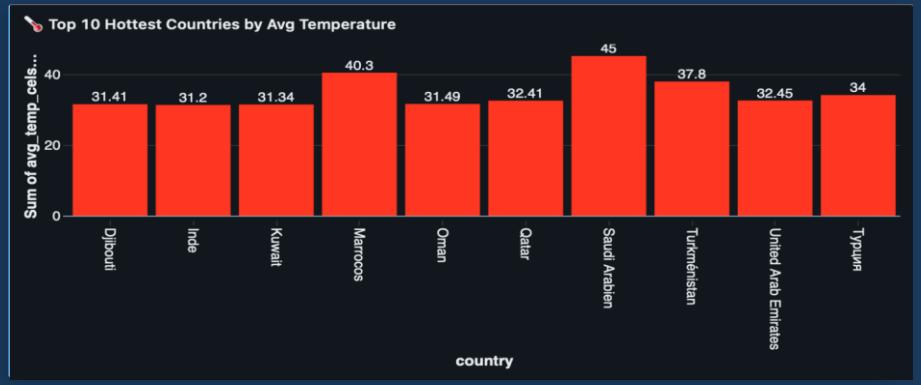
PM2.5, PM10, CO, Ozone, NO2, SO2 levels by region. US EPA & GB DEFRA index categories — Good/Moderate/Unhealthy/Hazardous zones.

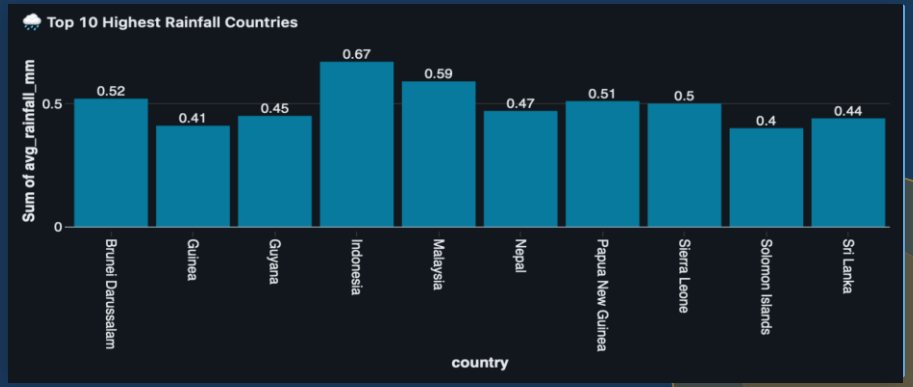
Global Weather Heatmap

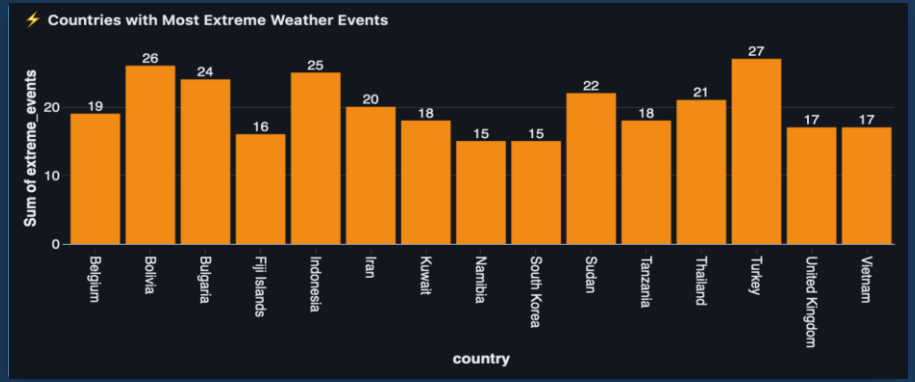
Multi-metric comparison across 100+ countries. Condition text frequency, cloud cover patterns, visibility categories, and sensor type distribution.

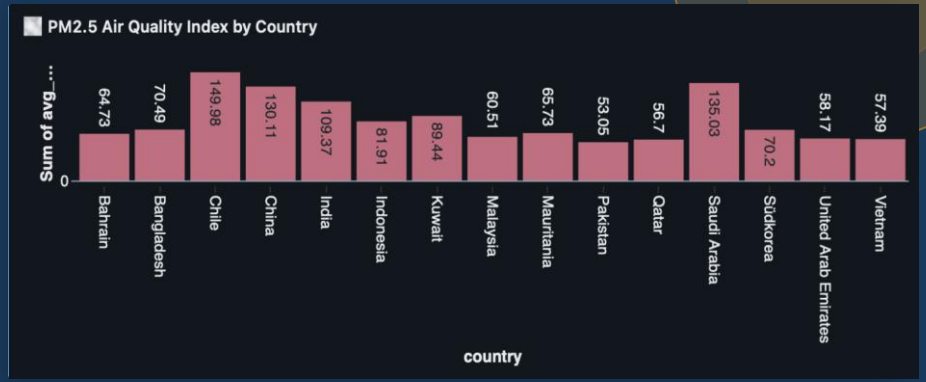
DASHBOARD SCREENSHOTS

Databricks dashboards built on Gold layer — live weather analytics









KEY LEARNINGS & CONCLUSION

Medallion Architecture

Bronze = immutable truth, Silver = trustworthy, Gold = business-ready. Each layer has a strict contract — never skip or merge layers.

Real-Time Streaming at Scale

S3 Event → Lambda → Kinesis → Firehose is a zero-code managed chain. 127K records streamed in ~256 batches — zero manual steps.

Security by Default

4 IAM roles with least-privilege, Unity Catalog for S3 governance, PAT tokens — security is architecture, not an afterthought.

Serverless Has Constraints

Databricks Serverless blocks fs.s3a configs. Unity Catalog external locations is the production-correct, more secure solution.

Silver Layer Is Non-Negotiable

Even clean datasets need Silver: type casting, dedup, outlier detection, schema enforcement. Gold is only trustworthy if Silver is rigorous.

Orchestration = Observability

11 Airflow tasks + Slack on every state change = full pipeline observability. You always know what ran, when, and whether it succeeded.